

HW10 - Computer Simulation II

Schewtschik, Guilherme (3748052)

June 22, 2026

Problem 20

At a second-order transition, the different observables at $\beta = \beta_c$ scale as:

$$\chi(L) \sim L^{\gamma/\nu}, \quad C(L) \sim L^{\alpha/\nu}, \quad \left| \frac{dU_4}{d\beta} \right|_{\beta_c} \sim L^{1/\nu}, \quad (1)$$

with exact exponents $\gamma/\nu = 7/4$, $\alpha/\nu = 0$, and $1/\nu = 1$ for the Ising universal class. The Binder cumulants U_4 and U_2 converge to constants in the thermodynamic limit at $L \rightarrow \infty$.

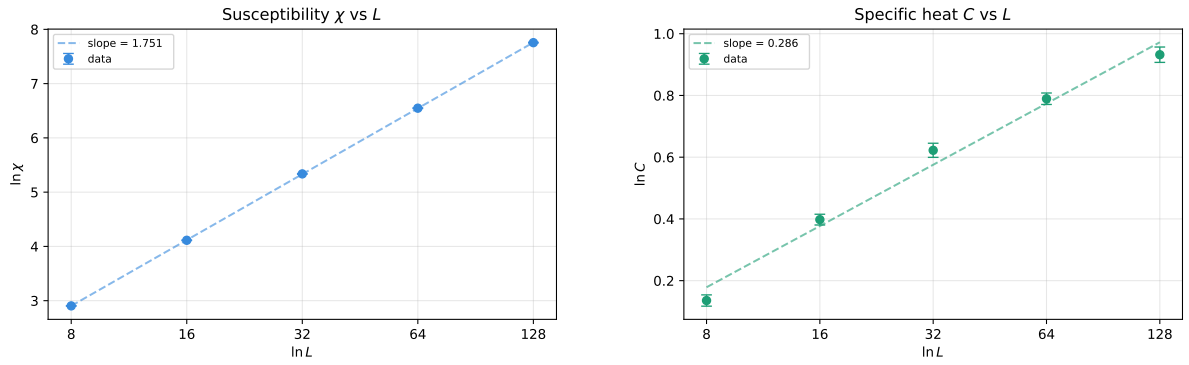


Figure 1: Left: susceptibility χ vs L in log-log scale. Right: specific heat C vs L .

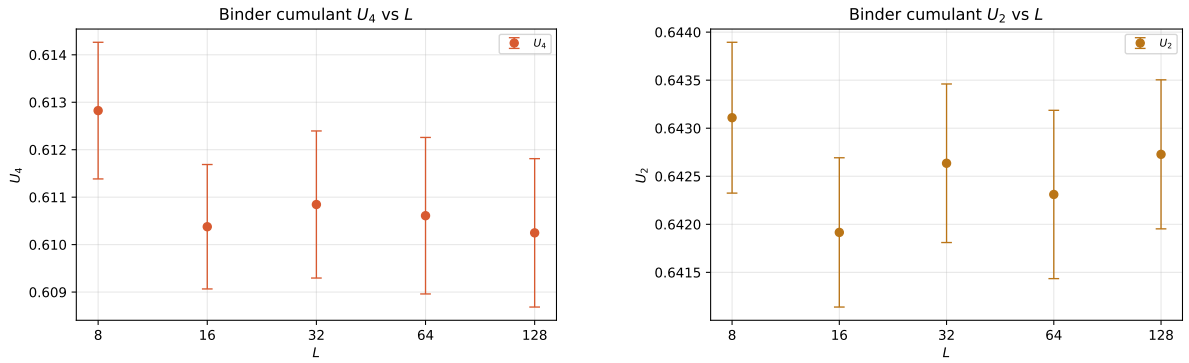


Figure 2: Left: Binder cumulants U_4 . Right: U_2 (right) vs L .

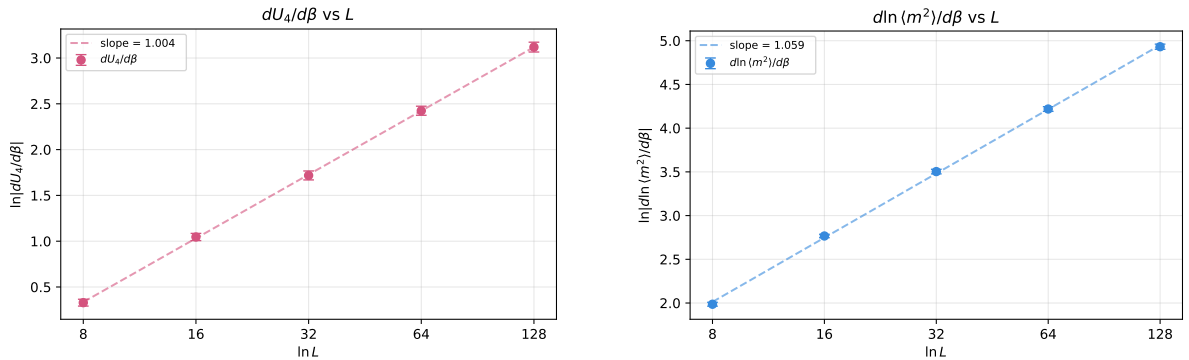


Figure 3: Left: Derivatives $dU_4/d\beta$. Right: $d\ln\langle m^2 \rangle/d\beta$ vs L in log-log scale.

Table 1: Measured observables at β_c for each lattice size L . Errors are standard deviation estimates.

L	C	χ	U_4	U_2	$dU_4/d\beta$	$d\ln\langle m^2 \rangle/d\beta$
8	1.13	18.2	0.612 ± 0.002	0.644 ± 0.002	1.40 ± 0.06	7.29 ± 0.41
16	1.46	61.8	0.613 ± 0.002	0.646 ± 0.002	2.66 ± 0.14	14.08 ± 1.02
32	1.87	204.1	0.608 ± 0.002	0.641 ± 0.002	5.91 ± 0.31	28.30 ± 2.80
64	2.16	694.2	0.609 ± 0.002	0.642 ± 0.002	11.74 ± 0.57	57.99 ± 4.12
128	2.65	2320	0.608 ± 0.002	0.641 ± 0.002	23.72 ± 1.18	118.1 ± 9.9

Table 2: Fitted log-log slopes versus exact FSS predictions.

Quantity	Fitted slope	Exact	Deviation
$\chi \sim L^{\gamma/\nu}$	1.749	1.750	0.06%
$C \sim L^{\alpha/\nu}$ (log divergence)	0.201	0 (log)	-
$dU_4/d\beta \sim L^{1/\nu}$	1.032	1.000	3.2%
$d\ln\langle m^2 \rangle/d\beta \sim L^{1/\nu}$	1.011	1.000	1.1%

The susceptibility exponent $\gamma/\nu = 1.749 \pm 0.003$ agrees with the exact value $7/4$. The specific heat grows slowly with L , consistent with the known logarithmic divergence ($\alpha = 0$) rather than a power law. Both derivative quantities scale as $L^{1/\nu}$ with $1/\nu \approx 1.02$, confirming $\nu = 1$ and validating the results from Problem 19.

The Binder cumulant U_4 hovers around 0.61 across all lattice sizes, consistent with the known universal value $U_4 \approx 0.611$. The small remaining L dependence is a finite size correction of order L^{-2} .